

Week 1 - Introduction to AI Engineering

Concepts:

AI fundamentals, Python basics

Project:

Build a simple calculator using Python

Outcome:

Understand the basics of AI and Python programming

Week 2 - Python for AI

Concepts:

NumPy, pandas

Project:

Work with datasets using NumPy and pandas

Outcome:

Understand how to work with data in Python

Week 3 - Machine Learning Fundamentals

Concepts:

Supervised learning, Unsupervised learning

Project:

Implement supervised and unsupervised learning algorithms

Outcome:

Understand the basics of machine learning

Week 4 - Deep Learning Fundamentals

Concepts:

Neural networks, Activation functions

Project:

Build a simple neural network

Outcome:

Understand the basics of deep learning

Week 5 - Natural Language Processing

Concepts:

Text preprocessing, Tokenization

Project:

Implement text classification using NLP

Outcome:

Understand the basics of NLP

Week 6 - Computer Vision

Concepts:

Image processing, Object detection

Project:

Implement image classification using computer vision

Outcome:

Understand the basics of computer vision

Week 7 - AI Ethics and Bias

Concepts:

Bias in AI, Fairness in AI

Project:

Analyze and mitigate bias in AI models

Outcome:

Understand the importance of AI ethics and bias

Week 8 - AI Project Development

Concepts:

Project planning, Project management

Project:

Develop a comprehensive AI project plan

Outcome:

Understand how to develop and manage an AI project

Week 9 - AI Model Deployment

Concepts:

Model deployment, Model serving

Project:

Deploy an AI model to a cloud platform

Outcome:

Understand how to deploy and serve AI models

Week 10 - AI Model Monitoring

Concepts:

Model monitoring, Model evaluation

Project:

Monitor and evaluate the performance of an AI model

Outcome:

Understand how to monitor and evaluate AI models

Week 11 - AI Model Maintenance

Concepts:

Model maintenance, Model updates

Project:

Maintain and update an AI model

Outcome:

Understand how to maintain and update AI models

Week 12 - AI Engineer Capstone Project

Concepts:

Project:

Develop a comprehensive AI project

Outcome:

Apply AI engineering skills to develop a real-world AI project
